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Ghosh(10) **Pub. No.: US 2021/0369545 A1**(43) **Pub. Date: Dec. 2, 2021**(54) **METHOD AND SYSTEM FOR SMART
NAVIGATION FOR THE VISUALLY
IMPAIRED**(71) Applicant: **Arko Ayan Ghosh**, Tampa, FL (US)(72) Inventor: **Arko Ayan Ghosh**, Tampa, FL (US)(21) Appl. No.: **16/842,706**(22) Filed: **Apr. 7, 2020****Publication Classification**(51) **Int. Cl.**

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ABSTRACT

In 2019, the World Health Organization stated that globally, approximately 2.2 billion people live with some form of vision impairment. Visual impairment limits the ability to perform everyday tasks and adversely affects the ability to interact with the surrounding world, thus discouraging individuals navigating unpredictable and unknown environments. The present invention is a method and a system to define and develop a smart navigation intelligent cane (i-Cane) that enables a visually impaired person to navigate his or her environment. The method and the system detects objects along the path of the visually impaired person, measures the distance of the objects from the person, identifies the objects, uses speech to alert the person of the approaching objects, the type of objects obstructing the path, and the distance between the objects and the person.

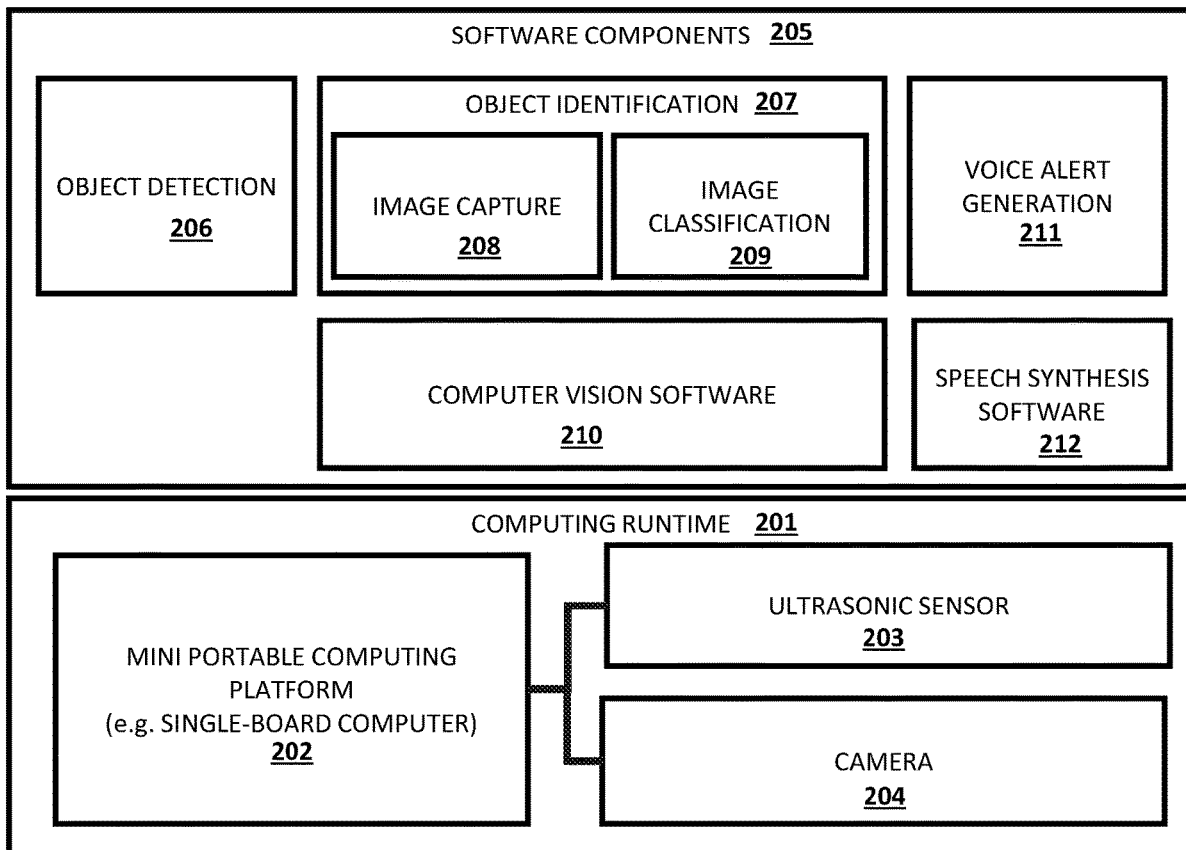
**Component Diagram 200**

Fig. 1

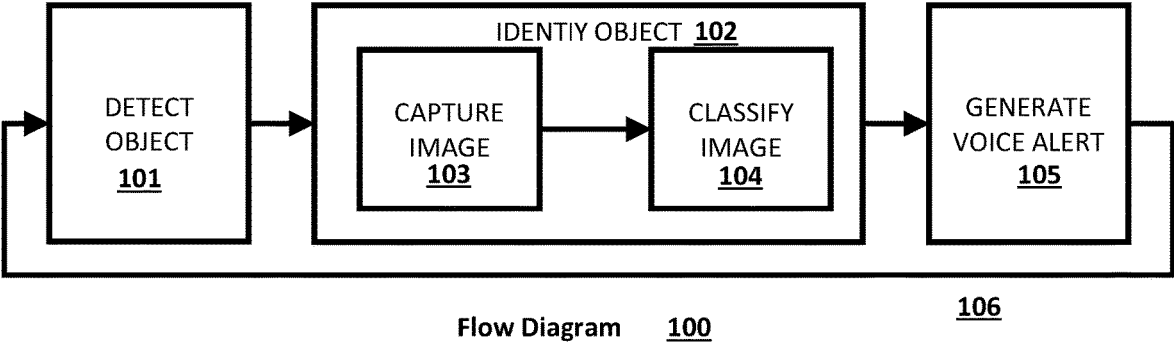
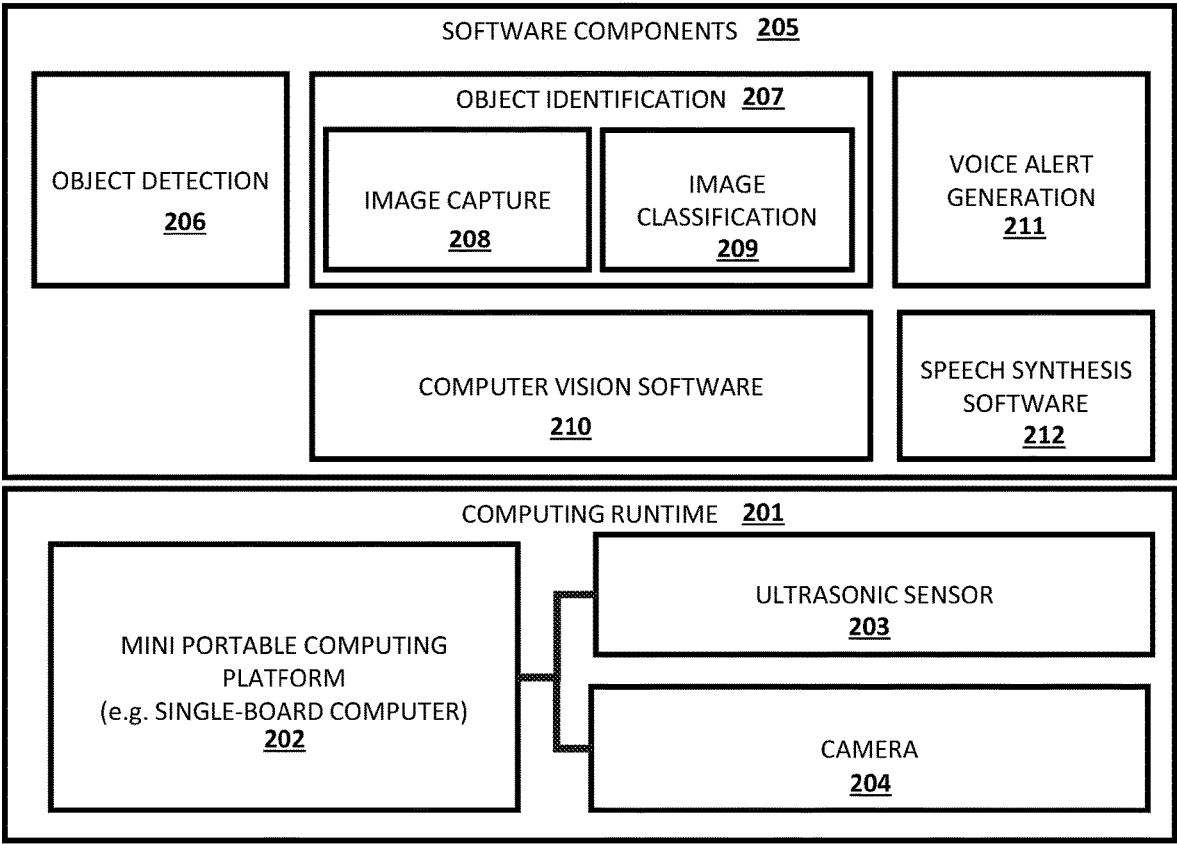
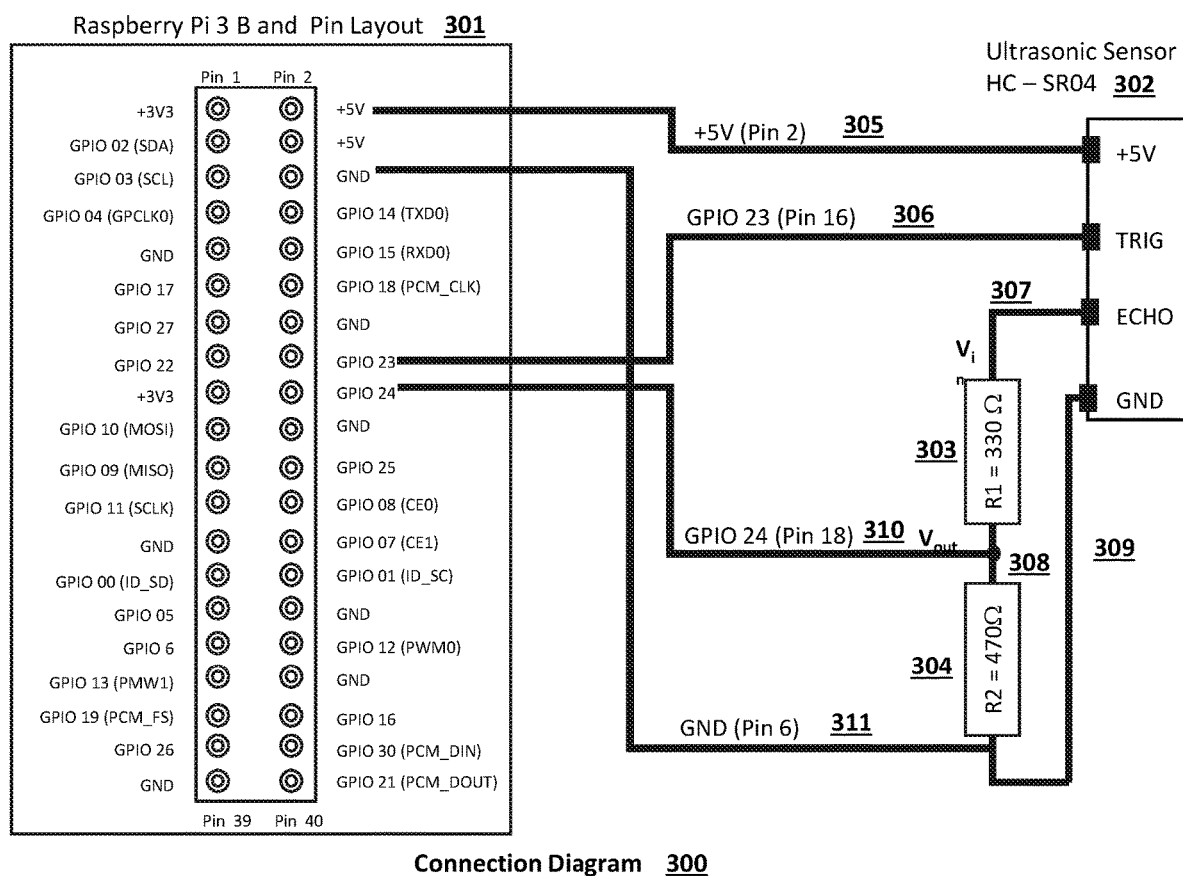


Fig. 2



Component Diagram 200

Fig. 3



METHOD AND SYSTEM FOR SMART NAVIGATION FOR THE VISUALLY IMPAIRED

FIELD OF INVENTION

[0001] The present invention is a method and a system to define and develop a smart navigation intelligent cane (i-Cane) that enables a visually impaired person to navigate his or her environment.

BACKGROUND OF THE INVENTION

[0002] In 2019, the World Health Organization stated that globally, approximately 2.2 billion people live with some form of vision impairment, of whom 1 billion people have moderate to severe vision impairment. Findings from the “Summary Health Statistics for U.S. Adults: National Health Interview Survey, 2012” established that an estimated 20.6 million adult Americans (or nearly 10% of all adult Americans in 2012) reported they either “have trouble” seeing, even when wearing glasses or contact lenses, or that they are blind or unable to see at all. Any form of visual impairment is severe enough to cause a significant impact on the course of their daily living. Specifically, their ability to move around and recognize obstacles may be compromised as they carry out their day-to-day life.

[0003] The world is full of dangers and wonders which are avoided or appreciated with our vision. The physical world poses the greatest challenge to the visually impaired person. How does one know what and where things are and how to obtain them? How does one understand where he/she wants to go without the danger of colliding with the objects around them?

[0004] Blind individuals may be discouraged from moving freely and comfortably. What can help them to identify the approaching objects in their path of navigation and determine how far these objects are from the individuals, when they are moving in a house or walking in a mall or strolling through isles in a grocery store?

[0005] Therefore, there is a need to define a method and a system to solve the problem and the challenge faced with visually impaired persons described above.

SUMMARY OF THE INVENTION

[0006] The present invention describes a method and a system for smart navigation for the visually impaired. The method defines an approach to develop a smart navigation intelligent cane (i-Cane) that enables a visually impaired person to navigate his or her environment. There are three main steps in this method:

[0007] detecting the approaching objects in the path of a visually impaired person using an ultrasonic sensor and calculating the distance of the objects from the visually impaired person carrying the i-Cane

[0008] identifying and classifying the approaching objects

[0009] generating voice alert indicating the type of the objects and the distance of the objects from the visually impaired person carrying the i-Cane warning the person of the approaching objects

[0010] The system to develop smart navigation for the visually impaired includes a computing runtime and the necessary software components. The computing runtime includes:

[0011] a mini portable computing platform, an ultrasonic sensor, and a camera

[0012] a set of software components that realize the method steps described above

[0013] The software components include:

[0014] object detection component

[0015] object identification component that in turn includes image capture component, classify image component, and computer vision component

[0016] voice alert generation component that includes speech synthesis component

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 illustrates the main process flow and steps for the method defined in this invention.

[0018] FIG. 2 depicts the system behind i-Cane and the underlying building blocks of computing runtime and software components.

[0019] FIG. 3 illustrates the connections between a mini portable computing platform such as the single-board computer Raspberry Pi 3, an ultrasonic sensor such as HC-SR04 and the circuitry to connect the sensor to the Raspberry Pi.

DETAILED DESCRIPTION OF THE INVENTION

[0020] Visual impairment has a severe impact on the course of daily living, discouraging individuals from moving freely in an unknown environment. The world is full of dangers and wonders which are avoided or appreciated with our vision. The physical world poses the greatest challenge for the visually impaired person. How does one know what and where things are and how to obtain them? How does one understand where he/she wants to go without the danger of colliding with things around them?

[0021] Blind individuals may be discouraged from moving freely and comfortably. What can help them to identify the approaching objects in their path of navigation and determine the distance of the objects from the individuals, when they are moving in a house or walking in a mall or strolling through isles in a grocery store?

[0022] The purpose of this invention is to define a method and a system to develop a simple but affordable way to assist visually impaired persons to navigate around their environment. The method defines an approach to develop a smart navigation intelligent cane (i-Cane) that aids a visually impaired person to move around the surroundings:

[0023] by first detecting the approaching objects in the path of the visually impaired person carrying the i-Cane, finding the distance between the approaching objects and the person, and then identifying the objects leveraging ultrasonic sensor, camera, and computer vision technology and

[0024] finally, by generating a voice/speech alert for the visually impaired person in natural language using speech synthesis technology

[0025] In FIG. 1, the Flow Diagram 100 shows the method developed in this invention and its overall flow and key steps. The key steps in this method are:

[0026] Detect Object (as shown by 101 in FIG. 1)—As the visually impaired person carrying the i-Cane travels through a path, first, detect the approaching object in

the path using an ultrasonic sensor and then calculate the distance between the object and the person carrying the i-Cane

[0027] Identify Object (as shown by **102** in FIG. 1)—Next, identify the object, if the distance between the approaching object and the visually impaired person carrying the i-Cane meets a certain distance threshold:

[0028] by capturing an image (as shown by **103** in FIG. 1) of the approaching object and

[0029] by classifying and labeling the image of the approaching object (as shown by **104** in FIG. 1) using computer vision technology

[0030] Generate Voice Alert (as shown by **105** in FIG. 1)—Finally, generate voice alert using speech synthesis technology to indicate the type of the object and the distance of the object from the visually impaired person carrying the i-Cane, forewarning the person about the approaching object in a natural language so that the person can take some corrective actions to avoid the potential collision with the object

[0031] Continue with the flow, as shown by **106** in FIG. 1, as the virtually impaired person continues with his/her path and as the objects appear in the path

[0032] As part of this invention, a system is also defined to demonstrate the method developed in this invention. FIG. 2 illustrates the Component Diagram **200** for the system that implements the method and its flow as depicted by **100** in FIG. 1. The system for designing and developing the i-Cane is composed of Computing Runtime (as shown by **201** in FIG. 2) and Software Components (as shown by **205** in FIG. 2). The Computing Runtime (as shown by **201** in FIG. 2) includes a single-board Mini Portable Computing Platform (as shown by **202** in FIG. 2) providing an execution environment for the software components implementing the method described above. The Computing Runtime (as shown by **201** in FIG. 2) also enables the single-board Mini Portable Computing Platform (as shown by **202** in FIG. 2) to interface with an Ultrasonic Sensor (as shown by **203** in FIG. 2) and a Camera (as shown by **204** in FIG. 2). A representative computing runtime for the defined system can be made of a Raspberry Pi 3 as the single-board mini portable computing platform, an HC-SR04 as the ultrasonic sensor, and a Pi Camera as the camera.

[0033] The Software Components (as shown by **205** in FIG. 2) includes:

[0034] Object Detection (as shown by **206** in FIG. 2) detects the approaching object in the path of the visually impaired person carrying the i-Cane using an ultrasonic sensor and calculates the distance of the object from the person carrying the i-Cane.

[0035] Object Identification (as shown by **207** in FIG. 2) identifies the object if the distance between the approaching object and the visually impaired person carrying the i-Cane is less than a certain distance threshold. The Image Capture sub-component (as shown by **208** in FIG. 2) within the Object Identification component takes a picture of the approaching object using the camera. And then, the Image Classification sub-component (as shown by **209** in FIG. 2) labels and classifies the image using the Computer Vision Software (as shown by **210** in FIG. 2) running on the cloud.

[0036] Voice Alert Generation (as shown by **211** in FIG. 2) generates a voice alert using the Speech Synthesis Software (as shown by **212** in FIG. 2) to inform the visually impaired person carrying the i-Cane of the approaching object, its type and the distance between the object and the person, forewarning the person of the approaching object.

[0037] In FIG. 3, the Connection Diagram **300** illustrates the connection between the Raspberry Pi 3 (the single-board mini portable computing platform) and HC-SR04 (the ultrasonic sensor) and the circuitry in between connecting the two hardware components. The Pi Camera is directly connected to the camera port on the Raspberry Pi 3 using a camera cable. The system consisting of the Raspberry Pi 3 mini portable computing platform connected with the HC-SR04 ultrasonic sensor and the Pi Camera is mounted on the i-Cane.

[0038] **301** in FIG. 3, Illustrates the single-board mini portable computing platform Raspberry Pi 3 and its pin layout. **302** in FIG. 3 shows the ultrasonic sensor HC-SR04 and its four pins, namely, 5V Power, TRIGGER (TRIG), ECHO, GROUND (GND).

[0039] Connecting Ultrasonic Sensor to Raspberry Pi 3

[0040] The 5V Power pin of the ultrasonic sensor is connected to the GPIO 5V pin (Pin number 2) of the Raspberry Pi 3 as shown by **305** in FIG. 3.

[0041] The TRIG pin of the ultrasonic sensor is connected to the GPIO 23 pin (Pin number 16) of the Raspberry Pi 3 as shown by **306** in FIG. 3.

[0042] The ECHO pin of the ultrasonic sensor is connected to the resistor R1 (330Ω, represented by **303** in FIG. 3) as shown by **307** in FIG. 3. The other end of the resistor R1 is connected to the resistor R2 (470Ω, represented by **304** in FIG. 3) as shown by **308** in FIG. 3.

[0043] The other end of resistor R2 is connected to GND pin of the ultrasonic sensor as shown by **309** in FIG. 3. And the common point of the resistor R2 and GND pin of the ultrasonic sensor is connected to GPIO GND pin (Pin number 6) of the Raspberry Pi 3 as shown by **311** in FIG. 3.

[0044] The common point of R1 and R2 resistors is connected to the GPIO 24 pin (Pin number 18) of the Raspberry Pi 3 as shown by **311** in FIG. 3. The GPIO 24 pin sits between the resistors R1 and R2 thereby forming a parallel circuit and hence, reducing the voltage to approximately 3V from 5V. Mathematically, $V_{out} = V_{in} \times R2 / (R1 + R2)$ where $V_{in} = 5V$, $R1 + R2 = 800 \Omega$ and $R2 = 470 \Omega$ and substituting the values V_{out} is approximately equal to 3V.

[0045] A software program using Python programming language is run on the Raspberry Pi 3 mini portable computing platform, as shown by **202** in FIG. 2 and as shown by **301** in FIG. 3, connected to the ultrasonic sensor, as shown by **203** in FIG. 2 and as shown by **302** in FIG. 3 through the circuitry as shown in FIG. 3 and connected to the Pi Camera, as shown by **204** in FIG. 2.

[0046] Object Detection component triggers a signal to the ultrasonic sensor then waits to receive the echo back from the sensor and calculates the distance between the ultrasonic sensor on the i-Cane and the approaching object using the formula:

$$S = 2D/t, \text{ therefore, } D = (S \times t) / 2$$

[0047] where,

[0048] S is Speed of sound, so $S=34030$ cm/s

[0049] D is Distance between the approaching object and sensor

[0050] t is Time taken for the sensor to receive the echo back

[0051] If the distance between the approaching object and the visually impaired person carrying the i-Cane is greater than a distance threshold value (e.g. 150 cm) the system does not attempt to identify the approaching object or generate a voice alert, continuing to detect the subsequent approaching object. The distance threshold value is configurable by visually impaired person.

[0052] Object Identification is composed of Image Capture and Image Classification sub-components. The Image Capture sub-component takes the picture of the approaching object using the Pi Camera. The Image Classification sub-component calls the Computer Vision Software component on a cloud platform to determine the label annotation of the image and classifies the image based on the labels with top relevancy scores.

[0053] Voice Alert Generation component generates an audio alert using the Speech Synthesis software indicating the type of the approaching object and the distance between the approaching object and the visually impaired person carrying the i-Cane, thereby alerting the person of the approaching object so that the person can take some corrective actions to avoid the potential collision with the object.

NON-PATENT CITATIONS

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1. A method to define and develop a smart navigation intelligent cane (i-Cane) that aids a visually impaired person to move around the surroundings, the method comprising of:

first, detecting the approaching objects along the path of the visually impaired person carrying the i-Cane using an ultrasonic sensor and then calculate the distance of the objects from the person carrying the i-Cane (Detect Object)

next, identifying the objects, if the distance between the approaching objects and the visually impaired person carrying the i-Cane meets a certain distance threshold (Identify Object):

by capturing an image of the approaching objects (Capture Image) and

by labeling and classifying the image of the approaching objects using computer vision technology (Classify Image)

finally, generating a voice alert using speech synthesis technology to indicate the type of the object and the distance between the object and the visually impaired person carrying the i-Cane, forewarning the person of the approaching object in a natural language (Generate Voice Alert)

and continuing the flow and repeating steps of object detection, object identification (image capture and classification), voice alert generation, as the virtually impaired person continues with his/her path and as the objects appear in the path.

2. A system for implementing and demonstrating the method, as described above, to define and develop a smart navigation intelligent cane (i-Cane) that enables a visually impaired person to navigate the environment, the system comprising of:

a computing runtime that consists of:

a single-board mini portable computing platform such as the Raspberry Pi 3, mounted on an intelligent cane, i-Cane, providing an execution environment for the software components implementing the method described above

an ultrasonic sensor such as HC-SR04 connected to single-board mini portable computing platform using a circuitry

a camera such as the Pi Camera connected to the camera port on the single-board mini portable computing platform using a camera cable

a software program implementing multiple software components that

detects the approaching object in the path of the visually impaired person carrying the i-Cane using an ultrasonic sensor and calculates the distance of the object from the person carrying the i-Cane

triggers signals to the ultrasonic sensor to measure the distance of the obstacle and then waits to receive the echo back from the sensor

calculates the distance between the ultrasonic sensor on the i-Cane and the approaching object using the formula

$$S=2D/t, \text{ therefore, } D=(S \times t)/2$$

where,

S is Speed of sound, so $S=34030$ cm/s

D is Distance between the approaching object and sensor

t is Time taken for the sensor to receive the echo back

continues to detect the subsequent approaching objects and doesn't attempt to identify the approaching object or generate a voice alert, if the distance between the approaching object and the visually impaired person carrying the i-Cane is greater than a distance threshold value that is configurable for a given person

captures an image of the approaching objects by interfacing with a camera such as the Pi Camera, if the distance between the approaching object and the visually impaired person carrying the i-Cane is less than the distance threshold value

identifies the objects by

calling the computer vision software passing the captured image

classifying the image based on the label annotations and the corresponding relevancy scores returned back from the computer vision software generates an audio alert using speech synthesis software indicating the type of the approaching object and the distance between the approaching object and the visually impaired person carrying the i-Cane, thereby alerting the person of the approaching object.

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